Drew University - CSCI 260 - Fall 2024
C Programming Project

Project Info.: Due on 12/13/2024 (F) by 11:59pm, 15% of the total grade
Submission: On Moodle by everyone; see below.

In this project, you are asked to implement a "Library Borrowing System" using the C programming language. The program is going to be an application of the data structure, Linked List, that we cover in this course. To implement a Linked List, you will need to utilize *struct* in C.

Description: The program works as such

1. In the *main()* function, we will be able to define Linked Lists that are named/identified by students' name. For example, we can have LLs, *John*, *Alice*, and *Bob*, all initialized to *NULL*. The LLs represent the "book borrowing records" of the students, correspondingly.
2. Thus, each node in the LL is an entry of such record. An entry contains three attributes (besides the "next record" attribute), namely, *book title*, *due date* (how many days until the return deadline), and *if reserved* (a char ['T'/'F'] indicates if the book is reserved by the student; hence, has no deadline).
3. For example, the LL *John* can be a list with 3 entries:

   ("Calculus I", 30, 'F') –> ("Calculus II", 40, 'F') –> ("Computer Architecture", 0, 'T') –> NULL

   where note that, since the last record is "reserved" ('T'), then the due date attribute is 0.
4. Then, the librarian can modify a LL, so a new "book borrowing record" entry can either be **added** or **appended** to the LL (i.e., a new book is now checked-out), or an existing "book borrowing record" can be **delete** from the LL (i.e., a book has been returned).
5. The rule for the **appending** function is simply that, if an entry contains 'T' as its "reserved" value, then, the entry is going to be appended to the end of the list. If there are other entries that are already "reserved", then a new one is going to be the very last entry (or *tail*) of this LL.
6. For example, **appendLL(&John, "Calculus III", 0, 'T')** will cause the LL to become:

   ("Calculus I", 30, 'F') –> ("Calculus II", 40, 'F') –> ("Computer Architecture", 0, 'T')

   –>("Calculus III", 0, 'T') –> NULL
7. The rules for the **adding** function are that,
   (a) all entries that contain 'F' as their "reserved" values will appear before (more close to the *head*) those with 'T' as their "reserved" values.
   (b) Then, among all those 'F' entries, the records are ordered based on their "due date" entry. Whichever entry has a smaller "due date" is going to go before those with larger ones.
   (c) In case of tied scenario, the one being added will appear as the last one among those with the same "due date" value.
8. For example, from Step 6, **addLL(&John, "Machine Learning", 15, 'F')** will go before *"Calculus I"*; and both **addLL(&John, "Discrete Math", 30, 'F')** and **addLL(&John, "AI", 35, 'F')** will go between *"Calculus I"* and *"Calculus II"*; and **addLL(&John, "Linear Algebra", 60, 'F')** will go between *"Calculus II"* and *"Computer Architecture"*.

9. Lastly, the rule for the **deleting** function is simply if the entry we are trying to delete is in a student's record, delete that record while keeping all other records intact.

10. For Example, from Step 6, **deleteLL(&John, "Computer Architecture")** will cause the LL to become:

("Calculus I", 30, 'F') –> ("Calculus II", 40, 'F') –> ("Calculus III", 0, 'T') –> NULL

Algorithms: Please follow the algorithm provided in the lecture slides and textbook.

How to proceed: Work in groups of 2 people and each will be in charge of implementing some functions. Everyone is going to need to submit a copy on Moodle.

Note that the *main()* function is mainly intended for the librarian; i.e., creating the initial LL for a student and modifying a student's record. Therefore, we will be using it for our testing purposes.

You should provide enough comments to indicate what each line/section of code is trying to achieve. A well-commented submission will received 2% extra credits.

Submission: You must submit a .c file that contains an compilable (can compile without errors), executable, and well-commented code for the project. Put group members names in the comment on the very top of that file. Optionally, you can submit a .pdf file containing any development documentations you want to share. By showing good effort via that document will earn you up to 2% extra credits.